

Supplementary Material

Marginal Matching Does Not License Factorized Sampling: Auditing Conditional Style Leakage in Factorized Generative Models

Anonymous submission

A. Proofs

A.1 Proposition 1 (No certificate)

Proposition 1. Let $z_s \sim \mathcal{N}(0, I)$ on $\mathbb{R}^{d_s}$, and let $\{A_k\}_{k=1}^K$ be any measurable partition of $\mathbb{R}^{d_s}$ with $P(z_s \in A_k) = 1/K$ for each k (such a partition always exists, e.g. via level sets of a continuous statistic of z_s). Define $y = k \iff z_s \in A_k$. Then (i) $q(z_s) = \mathcal{N}(0, I)$ exactly, so $D(q(z_s), \mathcal{N}(0, I)) = 0$ for every divergence D ; and (ii) $I(z_s; y) = H(y) = \log K$.

Proof. (i) y is defined post hoc as a measurable function of z_s ; the distribution of z_s itself is never altered, so $q(z_s) = \mathcal{N}(0, I)$ holds exactly, and every divergence to $\mathcal{N}(0, I)$, aggregate MMD included, is identically zero. (ii) Since y is a deterministic function of z_s , $H(y | z_s) = 0$, so $I(z_s; y) = H(y) - H(y | z_s) = H(y)$. By construction $P(y = k) = 1/K$ for all k , so $H(y) = \log K$, which is also the maximum possible entropy, and hence the maximum possible mutual information with any other variable, for a K -ary label. $\square$

The construction is deliberately adversarial: $\{A_k\}$ need not resemble anything a trained decoder produces, and $q(z_s | y=k)$ here is z_s restricted to A_k and renormalized, not a smooth family with class-specific means. That is the point. It shows that no divergence computed on $q(z_s)$ alone can rule out z_s being maximally class-informative, regardless of how the dependence is structured geometrically. It is a worst-case non-identifiability result, not a claim about typical behavior.

A.2 Theorem 1 (Factorized-sampling mismatch)

Theorem 1. Let y be uniform on $\{1, \dots, K\}$, let $q(z_c, z_s | y=k)$ be the encoder’s joint class conditional, let $p(z_c | y=k)$ and $p(z_s)$ be the semantic and style priors, and let $q(z_s) = \mathbb{E}_y[q(z_s | y)]$. Define

$$M_{\text{fact}} := \mathbb{E}_y[\text{KL}(q(z_c, z_s | y) \| p(z_c | y)p(z_s))].$$

Then

$$M_{\text{fact}} = I_q(z_c; z_s | y) + \mathbb{E}_y[\text{KL}(q(z_c | y) \| p(z_c | y))] \\ + I_q(z_s; y) + \text{KL}(q(z_s) \| p(z_s)),$$

and all four terms are non-negative.

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Proof. Fix k and expand the class- k term by inserting $\pm \log q(z_c | k) \pm \log q(z_s | k)$ into the integrand:

$$\begin{aligned} & \text{KL}(q(z_c, z_s | k) \| p(z_c | k)p(z_s)) \\ &= \mathbb{E}_{q(z_c, z_s | k)} \left[\log \frac{q(z_c, z_s | k)}{q(z_c | k)q(z_s | k)} \right] \\ &+ \mathbb{E}_{q(z_c | k)} \left[\log \frac{q(z_c | k)}{p(z_c | k)} \right] + \mathbb{E}_{q(z_s | k)} \left[\log \frac{q(z_s | k)}{p(z_s)} \right] \\ &= I_q(z_c; z_s | y=k) + \text{KL}(q(z_c | k) \| p(z_c | k)) \\ &+ \text{KL}(q(z_s | k) \| p(z_s)), \end{aligned}$$

where the first term is, by definition, the mutual information between z_c and z_s under $q(\cdot, \cdot | y=k)$. Taking $\mathbb{E}_y[\cdot]$ gives the first two terms of the statement directly, since $\mathbb{E}_y[I_q(z_c; z_s | y=k)] = I_q(z_c; z_s | y)$ by definition of conditional mutual information. For the remaining piece, $\mathbb{E}_y[\text{KL}(q(z_s | y) \| p(z_s))]$, insert $\pm \log q(z_s)$ and use $q(z_s) = \mathbb{E}_y[q(z_s | y)]$:

$$\begin{aligned} \mathbb{E}_y[\text{KL}(q(z_s | y) \| p(z_s))] &= \mathbb{E}_y[\underbrace{\text{KL}(q(z_s | y) \| q(z_s))}_{= I_q(z_s; y)} + \text{KL}(q(z_s) \| p(z_s))] \\ &= I_q(z_s; y) + \text{KL}(q(z_s) \| p(z_s)), \end{aligned}$$

where the first bracket is exactly $I_q(z_s; y)$: the average KL of a conditional from its own marginal is, by definition, the mutual information. Non-negativity of every term follows from non-negativity of KL divergence and of mutual information (itself a KL divergence). $\square$

A.3 Corollaries

Corollary 1 (Exact sampling validity). $q(z_c, z_s | y=k) = p(z_c | y=k)p(z_s)$ for almost every k if and only if $M_{\text{fact}} = 0$, i.e. iff all four terms vanish simultaneously. In particular, class-invariant style ($I_q(z_s; y) = 0$) is necessary but not sufficient.

Proof. Every term of Theorem 1 is a non-negative expectation of a KL divergence (terms 2 and 4 are mutual informations, themselves KL divergences), so their sum M_{fact} is zero iff each term is zero; and $\mathbb{E}_y[\text{KL}(\cdot \| \cdot)] = 0$ iff the integrand vanishes for a.e. y , iff the two per-class distributions coincide a.e. $\square$

Corollary 2 (Decoder pushforward bound). Let Dec be the decoder and $\text{Dec}_{\#}\mu$ the law of $\text{Dec}(z_c, z_s)$ when $(z_c, z_s) \sim \mu$. Then

$$\mathbb{E}_y[\text{KL}(\text{Dec}_{\#}q(\cdot | y) \| \text{Dec}_{\#}[p(z_c | y)p(z_s)])] \leq M_{\text{fact}}.$$

Proof. The data-processing inequality states that pushing two distributions through the same Markov kernel (here Dec) cannot increase their KL divergence; applying it inside $\mathbb{E}_y[\cdot]$ and invoking Theorem 1 gives the bound. $\square$

This runs one way only. A small M_{fact} suffices for faithful class-conditional generation, but a large one does not force a visible failure, because a decoder can be insensitive to the mismatched latent directions and map distinct latent distributions to nearly identical image distributions. Whether the trained decoder actually conditions on the leaked structure that terms (2) and (4) quantify is an empirical question the decomposition does not settle, which is why the latent-swap intervention in the main paper is necessary rather than redundant.

B. Architecture Details

The semantic head produces $(\tilde{\mu}_c, r_c)$, giving $\mu_c = \tilde{\mu}_c / \|\tilde{\mu}_c\|_2$ and concentration $\rho_c = \sigma(r_c)(1 - \varepsilon)$; the semantic latent is sampled by the Spherical Cauchy Möbius reparameterization $\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$, $z_c = T_{\mu_c, \rho_c}(\eta_c)$. The style head produces $(\mu_s, \log \sigma_s^2)$ and $z_s = \mu_s + \sigma_s \odot \varepsilon_s$, $\varepsilon_s \sim \mathcal{N}(0, I)$.

The ResNet-18 backbone for 32×32 inputs replaces the first 7×7 convolution (stride 2) with a 3×3 convolution (stride 1, padding 1) and removes the max-pool layer, yielding a 4×4 spatial output with 512 channels; a shared FC layer maps $512 \times 4 \times 4 = 8192$ to 256 dimensions with SiLU, from which separate linear heads produce the semantic and style parameters. The decoder concatenates $[z_c; z_s]$ ($d_c=64$, $d_s=128$ in all experiments unless stated) and maps it through three ResBlockUp stages (bilinear $2 \times$ upsample, 1×1 projection, GroupNorm+SiLU+Conv residual block; $512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ channels, $4 \times 4 \rightarrow 32 \times 32$), followed by a 3×3 output convolution and sigmoid.

For each class k , F-CS-WAE maintains a center $m_k \in \mathbb{S}^{d_c-1}$ with prior $p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p)$, $\rho_p=0.7$, updated by EMA from batch class-mean directions once per epoch, $m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau)\bar{\mu}_c^{(k)})$, $\tau=0.95$. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \delta(t)\mathcal{L}_{\text{style-cl}} + \eta(t)\mathcal{L}_{\text{cls}},$$

with $\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x})$, $\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_k \text{MMD}^2(Q_k, P_k)$ with $Q_k = \{z_c^i: y_i=k\}$ and $P_k \sim p(z_c | y=k)$, $\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}, \{z_{c,r_j}^p\})$ against a random-class prior mixture, and $\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y)$ a linear auxiliary classifier preventing semantic collapse.

C. Training Hyperparameters and Schedule

Per-class style regularization ramps up in Phase B, before $\mathcal{L}_{\text{class}}$ and $\mathcal{L}_{\text{agg}}$ are introduced in Phase C, so that it begins as early as possible relative to the semantic MMD terms, limiting rather than eliminating the epochs during which class-specific (z_c, z_s) interactions can form without style-conditional counter-pressure. Phase A itself (epochs 0–49) is reconstruction-only, so the decoder does see real (z_c, z_s)

Table 1: Hyperparameters used in all F-CS-WAE experiments.

Parameter	Value
Optimizer	Adam, lr = 10^{-3}
LR schedule	StepLR, step 100, $\gamma = 0.5$
Batch size	128
Total epochs	300
Gradient clip (global norm)	1.0
Semantic dim d_c / style dim d_s	64 / 128
Prior concentration ρ_p	0.7
EMA momentum τ	0.95
$\lambda_1 / \lambda_{\text{LPIPS}}$	1.0 / 0.1
α_f / β_f (class / agg, MMD)	2.0 / 5.0
γ_f (style)	1.0
δ_f (per-class style, off/on)	0 / 1
η_f (classifier)	0.3
Style prior temperature τ_s	0.25

Table 2: Training phase boundaries and active loss terms; ramps are linear.

Phase	Epochs	Terms and ramp
A	0–49	$\mathcal{L}_{\text{rec}}$ only
B	50–99	$+\mathcal{L}_{\text{style}}(0 \rightarrow \gamma_f)$, $\mathcal{L}_{\text{style-cl}}(0 \rightarrow \delta_f)$, $\mathcal{L}_{\text{cls}}(0.1 \rightarrow 0.2)$
C	100–199	$+\mathcal{L}_{\text{class}}(0 \rightarrow \alpha_f/2)$, $\mathcal{L}_{\text{agg}}(0 \rightarrow \beta_f/2)$, $\mathcal{L}_{\text{cls}}(0.2 \rightarrow 0.3)$
D	200–299	$\mathcal{L}_{\text{class}}(\alpha_f/2 \rightarrow \alpha_f)$, $\mathcal{L}_{\text{agg}}(\beta_f/2 \rightarrow \beta_f)$

pairs before any style or class regularization is active; Experiment 7 in the main paper tests whether removing this warmup changes the leakage, and finds it does not.

Baselines (ResNetAE, VAE, WAE-MMD, VaDE) use a unified lightweight CNN protocol, 50 epochs, Adam lr 10^{-3} , StepLR(30, 0.5); the matched-supervision baselines of Table 1 use the same ResNet-18 backbone and total latent dimension ($d=192$) as F-CS-WAE, trained for 300 epochs. The cross-model diagnostic baselines use a separate lightweight 3-layer CNN encoder trained for 100 epochs, which is why architecture and budget are not matched across the rows of Table 2 in the main paper.

D. Diagnostic Protocol

All leakage diagnostics use $n=2048$ held-out test samples with a fixed seed, so repeated invocations across checkpoints evaluate the same subset rather than adding sampling noise on top of single-seed training runs. The Euclidean MMD estimator uses a multi-scale RBF bandwidth ladder $\sigma \in \{0.5, 1, 2, 5, 10, 20, 50\}$, averaged; a single median-heuristic bandwidth saturates at z_s 's empirical scale and returns near-zero regardless of distributional mismatch. The same saturation is why we do not report the HSIC proxy: with a single fixed bandwidth it returns a near-constant value across checkpoints whose Δ_{inter} and LP vary by factors of five.

JointMMD (Eq. 3 in the main paper) uses a product kernel: the multi-scale spherical RBF above for z_c and the multi-scale Euclidean RBF for z_s , with per-block standardization. **The permutation null independently permutes both sample**

sets within each class, so any residual value reflects estimator bias rather than dependence; we report 200 permutations per checkpoint across 3 seeds.[†]

The linear probe is a single linear layer trained on the encoded μ_s of the evaluation split with an 80/20 train/validation split, Adam at 10^{-2} for 100 epochs; we report validation accuracy. Because the probe is linear, it lower-bounds the recoverable class information: a non-linear probe would only increase LP, so the values we report are conservative with respect to the leakage claim.

The latent-swap intervention draws $n=100$ pairs per ordered class pair (90 off-diagonal pairs for $K=10$), decodes $\text{Dec}(\mu_c, \mu_s)$ using posterior means rather than samples to remove reparameterization noise, and re-encodes the result. We report the diagonal-excluded means of $P(\text{pred}=a)$ (content-following) and $P(\text{pred}=b)$ (style-following); these do not sum to one, and the residual is the mass assigned to the other eight classes.

E. Extended Results

E.1 Full δ sweep

Table 3 gives the complete six-point sweep including reconstruction and sample-quality columns omitted from the main paper. Naive FID improves monotonically with δ on MNIST, from 74.0 at $\delta=0$ to 41.5 at $\delta=3$, which is worth stating plainly: the remedy makes naive prior samples *better* even though it does not make them correct. The two quantities are not the same thing, and conflating them is precisely the error the paper is about. Reconstruction is essentially flat across the sweep (SSIM 0.977–0.983 on MNIST), so the remedy is not purchasing leakage reduction with reconstruction quality.

Table 3: Full δ sweep, single seed per point (measured).

Data	δ	ACC	Δ_{inter}	LP	FID	SSIM
MNIST	0	.9915	6.21	99.9%	74.0	.983
MNIST	0.03	.9944	3.27	99.8%	62.8	.983
MNIST	0.1	.9898	2.12	99.3%	52.6	.983
MNIST	0.3	.9936	1.55	84.5%	59.4	.982
MNIST	1	.8592	1.22	42.6%	43.4	.979
MNIST	3	.9929	1.18	43.0%	41.5	.977
CIFAR	0	.8129	3.74	81.7%	80.5	.706
CIFAR	0.03	.8111	2.48	69.3%	79.1	.707
CIFAR	0.1	.8100	2.03	58.5%	77.5	.710
CIFAR	0.3	.7148	1.59	51.3%	72.0	.707
CIFAR	1	.8000	1.05	30.2%	74.4	.708
CIFAR	3	.7994	0.95	28.5%	76.1	.702

Note that the Δ_{inter} values here are computed on the δ -sweep checkpoints and differ in the third significant figure from the values in the main paper’s Table 2, which come from the separately-trained $\delta=0/\delta=1$ checkpoints used for the cross-model comparison (6.21 vs 6.27; 1.22 vs 1.46). This is ordinary run-to-run variation between independently trained models at the same setting, and is itself a small illustration of why the single-seed caveat in the main paper matters.

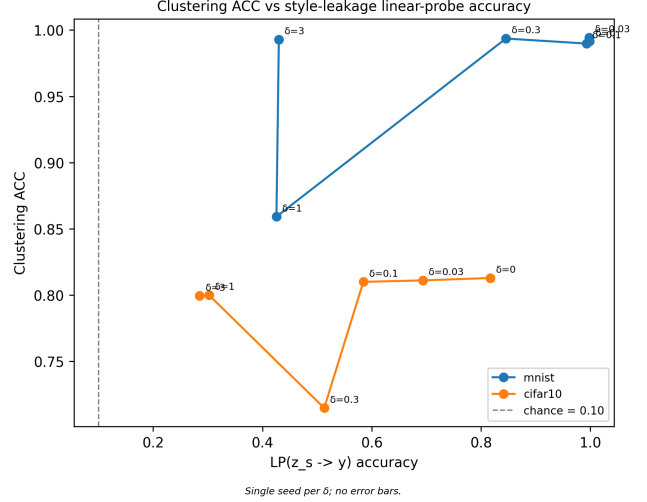


Figure 1: Clustering ACC against linear-probe leakage across the six-point δ sweep, both datasets (single seed per point; the connecting line is a visual aid, not a claim of a smooth underlying curve). Leakage (the x -axis) moves smoothly with δ ; ACC does not.

E.2 Cross-dataset leakage

Table 4: Leakage across three datasets at $\delta=0$ (measured), ordered by [measured[†]](#) intra-class diversity.

Dataset	Diversity [†]	Δ_{inter}	LP
MNIST	0.33 [†]	6.27	100.0%
Fashion-MNIST	0.47 [†]	4.86	97.7%
CIFAR-10	0.66 [†]	4.19	89.9%

Fashion-MNIST at $\delta=0$ reaches ACC 0.9341, NMI 0.8672, ARI 0.8633, FID 72.37, sitting between MNIST and CIFAR-10 on every axis (Figure 3). On MNIST the single-latent CS-WAE variant (one Spherical Cauchy semantic latent, no style variable) reaches ACC 0.8735, NMI 0.8863, ARI 0.8411, FID 26.95. It is worth noting explicitly that this ablation has *no style latent and therefore cannot leak by construction*, and it also achieves the best FID of any model we trained. That is not an argument for removing the style variable, since the factorization exists to provide independent style control, but it does bound how much the factorization is buying on this dataset.

E.3 Per-class conditional MMD

The single scalar Δ_{inter} compresses a K -way structure. Computing $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ per class at $\delta=0$ gives a mean of 0.0186 on MNIST against a global MMD of 0.00089, a factor of 21; on CIFAR-10, 0.0085 against 0.00056, a factor of 15. Every one of the ten per-class values on both datasets sits above the global figure, so this is not a few confusable classes dragging an average: every class contributes (Figure 4). At $\delta=1$ the ratios fall to $4.4\times$ (MNIST)

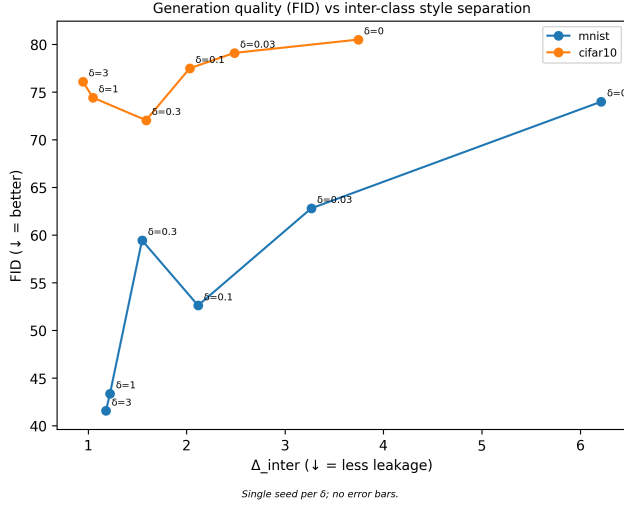


Figure 2: Naive-sampling FID against Δ_{inter} across the same sweep. Both fall together as δ grows: reducing conditional leakage also improves naive sample quality, even where it does not make sampling *correct*.

and $5.3\times$ (CIFAR-10) without reaching one. The pairwise structure $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$ likewise shows broad off-diagonal separation on both datasets rather than a few isolated pairs (Figure 5).

F. Failure-Mode Analysis

The naive-sampling failure is systematic rather than random. Confusion between the sampled class $y=k$ and the class the classifier assigns to the generated image (Figure 7) shows probability mass collapsing onto a small set of attractor digits, notably 8, largely independent of which class was requested. This is what one expects if the decoder resolves an out-of-training-distribution (z_c, z_s) pair by falling back on whichever class its style input most resembles, and it explains why self-accuracy sits near 0.15 rather than near the 0.10 that uniform random errors would produce: the errors are concentrated, not spread.

The latent-swap grid makes the same point at the pixel level (Figure 8). At $\delta=0$ every row (fixed z_c donor) produces visually identical output; only the column (z_s donor) determines the digit. On CIFAR-10 the rows are more similar to each other than in the MNIST grid but still show column-driven drift in object identity for several classes, consistent with a decoder that was never as exclusively dependent on z_s . The CIFAR-10 swap heatmaps (Figure 9) show the same $\delta=0 \rightarrow \delta=1$ flip as MNIST but weaker in magnitude (content-following $4.2\% \rightarrow 20.0\%$, style-following $68.8\% \rightarrow 38.5\%$).

A 2D t-SNE of z_s colored by class, across all six rows of the cross-model table (Figure 10), is worth reading with a caution attached: unlike a typical clustering figure, *color-separated clusters here mean worse*. Separation is visually obvious only for WAE-MMD and F-CS-WAE without the remedy, the two largest- Δ_{inter} rows, and visibly reduced af-

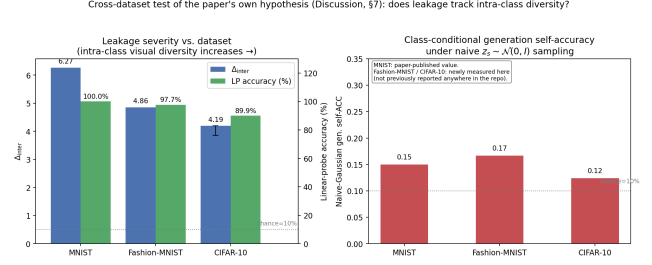


Figure 3: Left: Δ_{inter} and linear-probe accuracy across the three datasets, ordered by intra-class visual diversity; both decrease monotonically. Right: naive-Gaussian generation self-accuracy does *not* follow the same order (CIFAR-10 is lowest, not highest), because it additionally inherits dataset-intrinsic classification difficulty; see the main paper’s discussion.

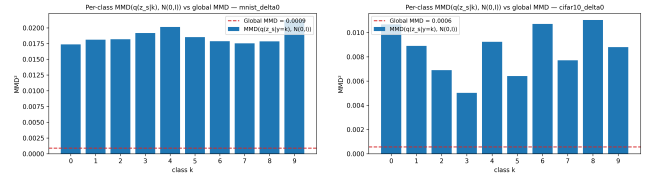


Figure 4: Per-class $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ at $\delta=0$ (bars) against the global MMD the marginal regularizer actually optimizes (dashed line). Left: MNIST; right: CIFAR-10. Every class sits far above the line the model was trained to minimize.

ter adding per-class style MMD. For VAE, β -TCVAE, and FactorVAE the leakage is statistically real (LP 74–87%) but not visually apparent in the projection. Projecting z_s onto the top principal component of the $K \times d_s$ class-mean matrix, the same signal Δ_{inter} summarizes, separates some classes cleanly even where t-SNE does not (Figure 6). A fixed low-dimensional projection can therefore both overstate and understate leakage depending on which directions it happens to preserve, which is the argument for the quantitative diagnostics over any single plot.

G. Sampling Strategies: Qualitative Results

Figures 11–14 give the qualitative side of the main paper’s sampling table. Two details are visible in the grids that the numbers compress away. First, the MNIST global-Gaussian samples are not noise: they are clean, legible digits of the *wrong* class, which is exactly the signature of a decoder resolving an unfamiliar (z_c, z_s) pair by trusting the style code. Second, the CIFAR-10 $\tau=0.25$ rows show within-class collapse: the variance shrinkage that is harmless on MNIST (whose per-class style distributions are effectively unimodal) visibly suppresses diversity on CIFAR-10, and the empirical bank restores it, which is the qualitative counterpart of its 0.241 diversity score.

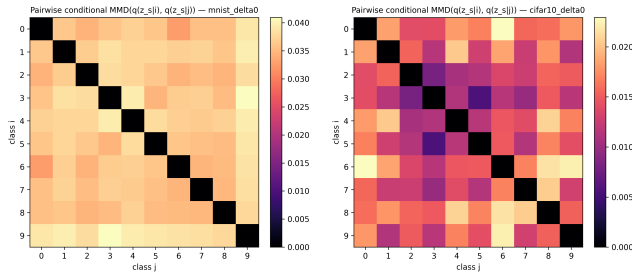


Figure 5: Pairwise $MMD^2(q(z_s | y=i), q(z_s | y=j))$ between all class pairs at $\delta=0$ (left MNIST, right CIFAR-10). Broad off-diagonal structure, not a few isolated confusable pairs.

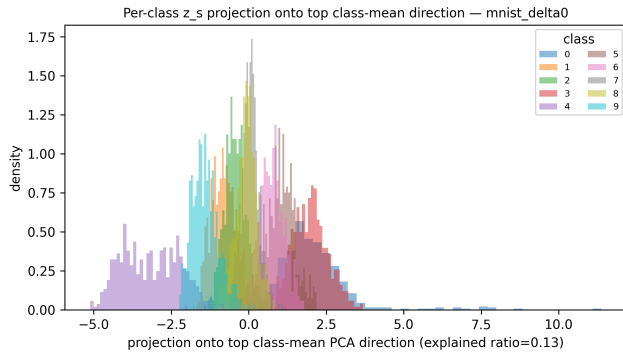


Figure 6: Per-class histograms of z_s projected onto the top principal component of the class-mean matrix (MNIST, $\delta=0$). Some classes separate cleanly along this single direction even where a 2D t-SNE does not make the separation visible, illustrating that a fixed low-dimensional projection can either overstate or understate leakage.

H. Case-Study Reference Results

Figures 15–16 document the case-study model’s baseline competence: stable clustering across seeds and recognizable class-conditional samples. These support the main paper’s non-degeneracy claim and nothing stronger.

I. Baseline Implementation

The four cross-model diagnostic baselines (VAE, WAE-MMD, β -TCVAE, FactorVAE) share a lightweight 3-layer convolutional encoder with latent dimension 64, trained 100 epochs with Adam at 10^{-3} on MNIST. β -TCVAE uses a minibatch-weighted-sampling estimate of the total correlation term; FactorVAE uses a density-ratio discriminator on permuted latent dimensions. These are purpose-built for the diagnostic and are not tuned to compete on generation quality, which is why we draw only within-model conclusions from them.

The seven matched-supervision baselines in the main paper’s Table 1 share F-CS-WAE’s ResNet-18 backbone and total latent dimension ($d=192$), trained 300 epochs under the same optimizer settings. They divide by how the label enters. *Label-guided* models put the label on the latent: AEWWithCE

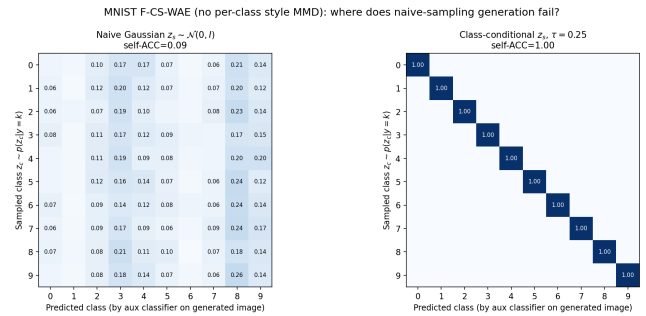


Figure 7: Confusion between the sampled class and the class assigned to the generated image. Naive Gaussian sampling (left) collapses onto attractor digits (notably 8) regardless of the intended class; the class-conditional prior (right, $\tau=0.25$) is a clean diagonal.



Figure 8: Latent-swap grids at $\delta=0$: row = z_c donor class, column = z_s donor class. Left, MNIST: every row is visually identical; identity is set entirely by the column. Right, CIFAR-10: weaker but still visible column-driven drift.

adds a cross-entropy head, AEWWithSupCon a supervised contrastive loss on L2-normalized projections, AEWWithCenterLoss a learnable per-class center pull, AEWWithTriplet an on-line hard-mined triplet margin loss. *Conditional-generative* models put the label only on the decoder: ConditionalVAE concatenates a label embedding to the decoder input, ConditionalWAE-MMD does the same with an MMD rather than a KL penalty, and GaussianClassPriorWAE matches per-class Gaussian priors in a Euclidean latent. The clustering gap between the two groups (up to 66.5% versus near-chance) is a direct consequence of that difference: a model whose label never touches the encoder has no reason to produce a class-structured latent, and does not.

J. Compute and Reproducibility

A single 300-epoch F-CS-WAE run takes roughly 4.5–13 hours wall-clock on one A30, the spread driven by contention from other jobs on the same node rather than by the configuration. Diagnostics run in seconds on a saved checkpoint. The full set of results in the main paper corresponds to approximately 30 training runs.

Every diagnostic is computed from a saved checkpoint

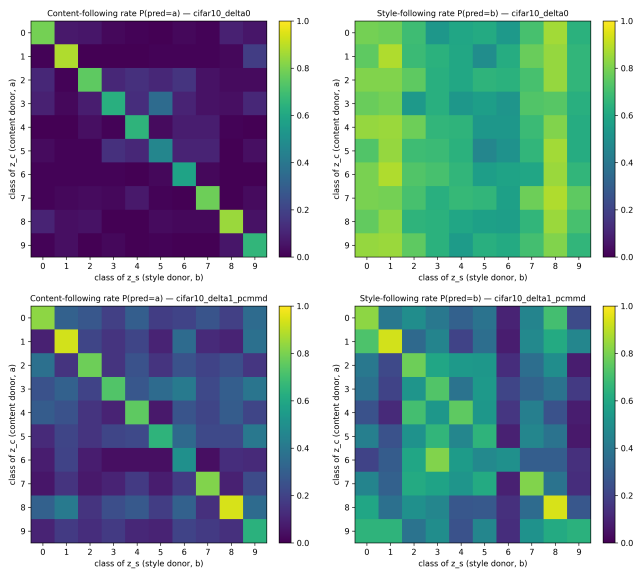


Figure 9: CIFAR-10 latent-swap heatmaps (left content-following, right style-following; top $\delta=0$, bottom $\delta=1$), the counterpart of the MNIST figure in the main paper.

with a fixed evaluation seed, so the numbers can be regenerated without retraining. The projected results marked $\dagger$ in this draft correspond to experiments that have not been run; of these, the sampling controls, the JointMMD permutation null, the support-graded swap, and the intra-class diversity measurement all operate on existing checkpoints and require no training, while the external classifiers, the multi-seed replication, and the invariance baselines of Table 3 require new runs.

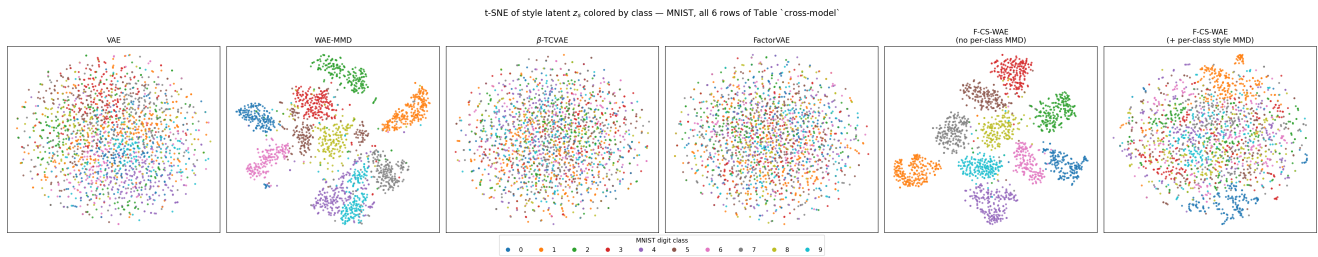


Figure 10: t-SNE of the style latent z_s , colored by MNIST digit class, for all six rows of the cross-model table. Color-separated clusters here mean *worse* (class information leaking into z_s): separation is visually obvious only for WAE-MMD and F-CS-WAE without the remedy, and visibly reduced after adding per-class style MMD; for the other three families the leakage is statistically real but invisible in this projection.

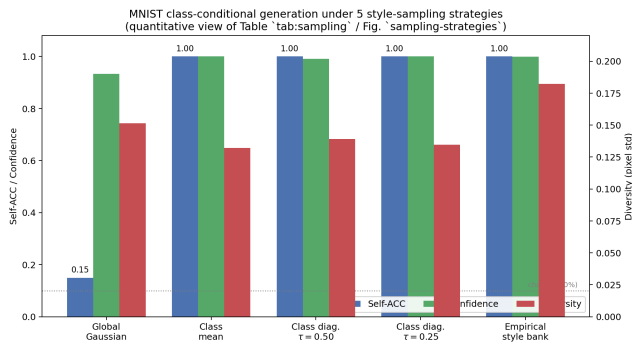


Figure 11: MNIST sampling strategies plotted directly: self-accuracy jumps from near-chance to 100% the moment z_s is conditioned on class in any way, while diversity (right axis) is lowest for the class-mean point estimate and highest for the empirical style bank.

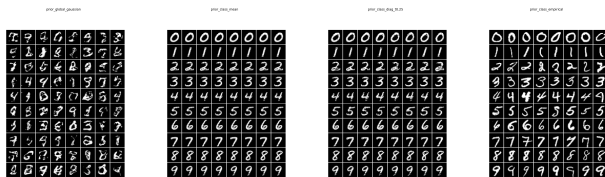


Figure 12: MNIST generation under four style-sampling strategies (left to right): global Gaussian (15% self-accuracy), class mean, class-conditional diagonal $\tau=0.25$, empirical bank (all 100%). The global Gaussian produces legible but frequently wrong-class digits; every class-conditional strategy restores the intended identity.



Figure 13: CIFAR-10 naive (odd rows) vs. class-conditional $\tau=0.25$ (even rows) samples, four classes. Diversity within a row collapses under $\tau=0.25$, and object identity is often not clearly recognizable by eye even where the classifier scores it correctly – the qualitative basis for the caution attached to the 0.681 number in the main paper.

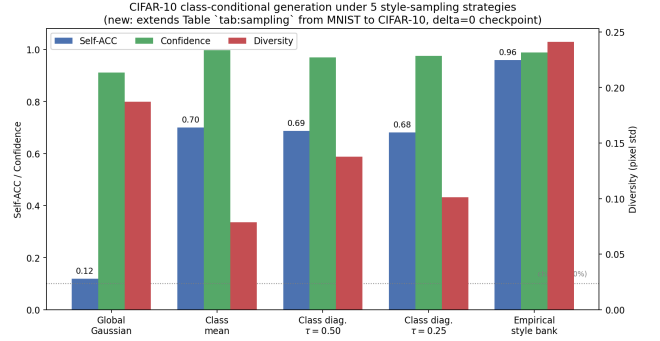


Figure 14: CIFAR-10 analogue of Figure 11: the class-conditional Gaussian strategies that reach 100% on MNIST plateau near 70% here; only resampling real encoded style codes (empirical bank) approaches MNIST-level self-accuracy.

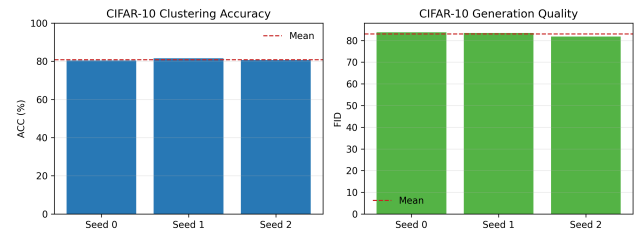


Figure 15: CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low (ACC std 0.52 points), which is why the main paper reports the 3-seed mean for F-CS-WAE while the baselines remain single-seed.

F-CS-WAE Class-Conditional Prior Samples: rows = class, columns = style samples

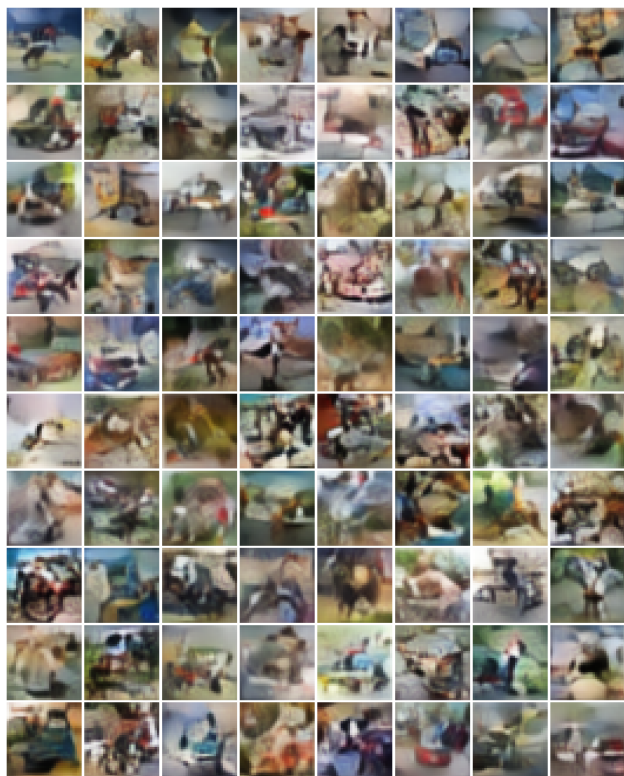


Figure 16: CIFAR-10 class-conditional prior samples (one row per class). FID 83 places these in the recognizable-but-blurry regime typical of L1/LPIPS-trained decoders without an adversarial loss, well short of photorealistic.